

TOLULOPE MATTHEW BUSOYE

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Professional Summary

Academically driven Electrical and Electronic Engineering graduate—best graduating student of the department and Faculty of Technology at the University of Ibadan. Strong technical and prototyping skills with a deep passion for research in embedded systems, edge AI, and intelligent energy management. Natural leader with proven ability to mentor students and deliver impactful engineering projects.

Interests: Robotics · Cyber-Physical Systems · Edge AI · Embedded Systems · Internet of Things · Computer Architecture · Intelligent Energy Management · Renewable Energy

Education

Master of Science (M.Sc.) in Mathematical Sciences (AI for Science Stream) 2026 – 2027
African Institute for Mathematical Sciences (AIMS) South Africa / Google DeepMind

- Theoretical knowledge and practical skills in mathematics, machine learning, and AI for scientific applications.

B.Sc. Electrical and Electronic Engineering 2019 – 2025
University of Ibadan, Nigeria CGPA: 3.94/4.00 — First Class

- Best Graduating Student of the Department and Faculty of Technology
- Studied electronic circuits, microprocessors and embedded systems, electric machines, programming, and cross-disciplinary engineering courses
- **Thesis:** Development of a Commercialised Solar-Based Mobile Energy Hub

Senior Secondary Certificate Examination 2015 – 2017
Pacesetters Comprehensive College

- Ranked among the top 5%; received the best student award in Physics, Mathematics, and Economics

Research Experience

Lead Student Investigator 2024 – 2025
Thesis: Development of a Commercialised Solar-Based Mobile Energy Hub
Supervisor: Prof. O. O. Olakanmi [Paper Link](#)

- Reviewed relevant literature, developed a research proposal, and created presentation materials
- Designed circuit schematics and prototyped a mobile solar-based energy hub, interfacing multiple microcontrollers, sensors, and actuators
- Integrated hardware and software systems; wrote firmware in C/C++ and mobile application in Flutter/Dart

Publications and Articles

- Busoye T. M., Jokomba Q. A., Busoye O. M., Joleto A. O. **Edge-Deployed Machine Learning Senosr (MLSenosr) for Real-Time Anomaly Detection in Electrical Submersible** *SPE Nigeria Annal International Conference and Exhibition, 2026*. DOI: [10.2118/235027-MS](#)
- Busoye O. M., Busoye T. M., Eigbe E. **Deep Microcompression: Structured Pruning and Bit-packed Quantization for Microcontrollers**. *PMLR Vol. 306 — ICML 2026 Global South in AI Workshop.*
- **Development of a Commercialised Solar-Based Mobile Energy Hub (Undergraduate thesis, University of Ibadan)**. (June 2025). DOI: [10.13140/RG.2.2.10441.86886](#)
- Inside TensorFlow Lite Micro on RP2040/RP2350: Setup and Project Structure Explained. (Sep 2025) Medium. [Link](#)
- Busoye, T. (in preparation). *A Compile-Time and Memory-Efficient Approach to SunSpec Models with FlatBuffers and Direct Modbus Register Mapping*. Draft manuscript.

Work Experience

Embedded Systems Engineer Jun 2025 – Present
Powerstart Integrated Technologies Limited (PowerLabs), Lagos

- Research and evaluate sensors for integration with Powerstart Autonomous Infrastructure (PAI) devices
- Implement SunSpec protocol to enable communication with SunSpec-compliant equipment
- Develop a hardware testing framework and integration-testing workflow supporting test-driven development
- Create and maintain a coding style guide to standardise hardware-related software development

Embedded Systems Intern

Powerstart Integrated Technologies Limited (PowerLabs), Lagos

Jan 2024 – Jun 2024

- Designed and developed a PAI device for power event capture based on Raspberry Pi Pico MCU
- Collaborated on software components for transmitting power events to Amazon AWS cloud endpoints
- Designed schematics and PCBs for Power Stool, a Bluetooth-controllable desk

Hardware & Electronics Intern

Reeddi Technologies, Lagos

Jun 2023 – Sep 2023

- Repaired 50 Reeddi Capsules, Energy Boxes, and Big Energy devices
- Designed a charge station for 100 Reeddi Capsules and set up Big Energy installations
- Developed device tracking systems

STEM Tutor & Engineer

Switches Technology, University of Ibadan

2021 – 2024

- Tutored and trained over 10 students in Arduino, Raspberry Pi, and embedded systems
- Designed innovative projects in embedded systems, IoT, and AI—including a Smart Distribution Board, Smart Bin, and Responsive Prosthetic Leg
- Oversaw 3D printing projects including a Head Mount Mouse and a 6-DOF Robotic Arm

Electrical & Electronic Engineering Intern

Crittall-Hope Nigeria Limited

Apr 2022 – May 2022

- Maintained single and three-phase machines; contributed to roller shutter production

Electrical & Electronic Engineering Intern

God's Will Technology Concept

Aug 2020 – Mar 2021

- Installed solar and inverter systems; performed electrical installations and electronic repairs
- Received foundational training in electrical systems and electronic repair techniques

Teaching Experience

Academic Tutor

Faculty of Technology, University of Ibadan

2021 – 2023

- Tutored over 70 fellow students in multiple courses including Elementary Computer Programming (FORTRAN)
- Tutored over 100 first-year students in PHY103 and MAT121

Academic Tutor

Toppers Tutorial, University of Ibadan

2021 – 2022

- Tutored over 100 first-year students in PHY103 and MAT111

Academic Tutor

Mountain of Fire and Miracle Ministries Campus Fellowship, UI

2021 – 2025

- Prepared academic materials and organised tutorials for colleagues and junior students

Leadership and Volunteering

President, Society of Electrical & Electronic Engineering Students (SEEES)

University of Ibadan

2023/2024

- Responsible for over 270 electrical and electronic engineering students
- Organised an industrial visit to Egbin Power Plc, Nigeria's largest power-generating station
- Coordinated workshops bridging theoretical knowledge and industry-required practical skills
- Increased financial resources for the next administration by over 2,000%

Lead & Co-Lead, Hardware & IoT, Google Developer Student Club (GDSC)

University of Ibadan

2021/22 – 2022/23

- Organised technical sessions and hardware workshops for over 50 students
- Served as Technical Lead, TechQuest GDSC UI 2023

Public Relations Officer, SEEES

University of Ibadan

2021 – 2023

- Led publicity and communication for all society activities; served as official spokesperson

Student Representative Council Member, 10th Assembly

Student Union, University of Ibadan

2022/2023

- Represented 100 students in constituency; responsible for policy-making

- Member, UISU House Committee on Audit and Electoral Matters

Scholarships and Awards

- **3rd Place**—Africa Energy Challenge, SPE Africa Annual Student Congress Apr 2025
- **1st Place**—SPE Energy Challenge, Student Technical Symposium & Exhibition, Nigeria Mar 2024
- NNPC/TotalEnergies National Merit Scholarship 2021 – 2024
- Federal Scholarship Board (FSB), Nigeria 2021 – 2023
- SPE Lagos Section Male Undergraduate Scholarship (Engineering) 2023
- Agbami Medical and Engineering Professionals Scholarship Award 2019 – 2022
- Most Outstanding in Academics—University of Ibadan Student Union 2023/24
- 5× Dean's Honour List—Faculty of Technology 2018/19 – 2023/24
- Most Outstanding Member—Junior Chamber International Nigeria, UI 2022/24
- 4× Academic Excellence Awards—MFMCF UI 2020/21 – 2023/24
- Silver Medal (1500m), Bronze Medal (400m)—TESA Dean's Cup 2021/22

Workshops and Conferences

- Adapting to Change, Edinburgh Earth Initiative, University of Edinburgh Nov 2024
- Nigeria Annual International Conference and Exhibition, SPE Nigeria Aug 2025
- SPE Africa Annual Student Congress, Society of Petroleum Engineers, Africa Apr 2025
- Student Technical Symposium & Exhibition, SPE Nigeria Mar 2024
- Student Professional Awareness Conference, IEEE, University of Ibadan Jul 2023
- Exhibitor, Energy Searchlight 2.0, Energy Club, University of Ibadan Feb 2025
- Exhibitor, Ambizero Conference, Energy Club & SPE, University of Ibadan Jun 2024
- Artificial Intelligence and Clean Energy (AI4CE) Workshop, First Technical University 2021

Selected Projects

- Automated Solar Power Trading and Distribution Unit
- Hybrid Solar/Grid Charging Station for 100 Reeddi Capsules
- STM32 PCB Design and Fabrication
- PCB Design for Smart Desk & Power Stool (PowerLabs)
- Automatic Solar Powered Traffic Light (RAINigeria)
- Smart Energy Monitoring and Distribution Board (SPE, STSE)
- Bluetooth Programmable Display (Arduino + Flutter)
- Mobile Controllable Robot Car · Responsive Prosthetic Leg · Wireless Controllable Dustbin System
- Assembly Language Projects: LED Counter, Binary Counter, LCD Control
- Over 15 Solar Installations

Skills and Languages

Technical: Circuit design, PCB design, Solar and inverter installation, Electronic prototyping
Software: C/C++, Python, KiCad, Docker, Git, Flutter/Dart, L^AT_EX, Microsoft Office, Canva
Strengths: Fast learning, Communication, Leadership, Time management, Adaptability, Creativity
Languages: English (fluent), Yoruba (native)

Professional Affiliations

- Graduate Member, Nigerian Society of Engineers (GMNSE)
- Student Member, Society of Petroleum Engineers (SPE)
- Student Member, Institute of Electrical and Electronics Engineers (IEEE)
- Member, Society of Electrical and Electronic Engineering Students (SEES), UI
- Member, Energy Club, University of Ibadan
- Student Member, Google Developer Student Club (GDSC)
- Member, Junior Chamber International Nigeria (JCIN)